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$$\begin{aligned} & \int \sin(x) \tan^2(x) dx \\ &= \int \sin(x) (\sec^2(x) - 1) dx \\ &= \int \frac{\sin(x)}{\cos^2(x)} dx - \int \sin(x) dx \end{aligned}$$

Substitueer $u = \cos(x)$

$$\begin{aligned} du &= -\sin(x) dx \\ &= \int \frac{-du}{u^2} - (-\cos(x)) + C \\ &= \frac{1}{\cos(x)} + \cos(x) + C \\ &= \sec(x) + \cos(x) + C \end{aligned}$$

References